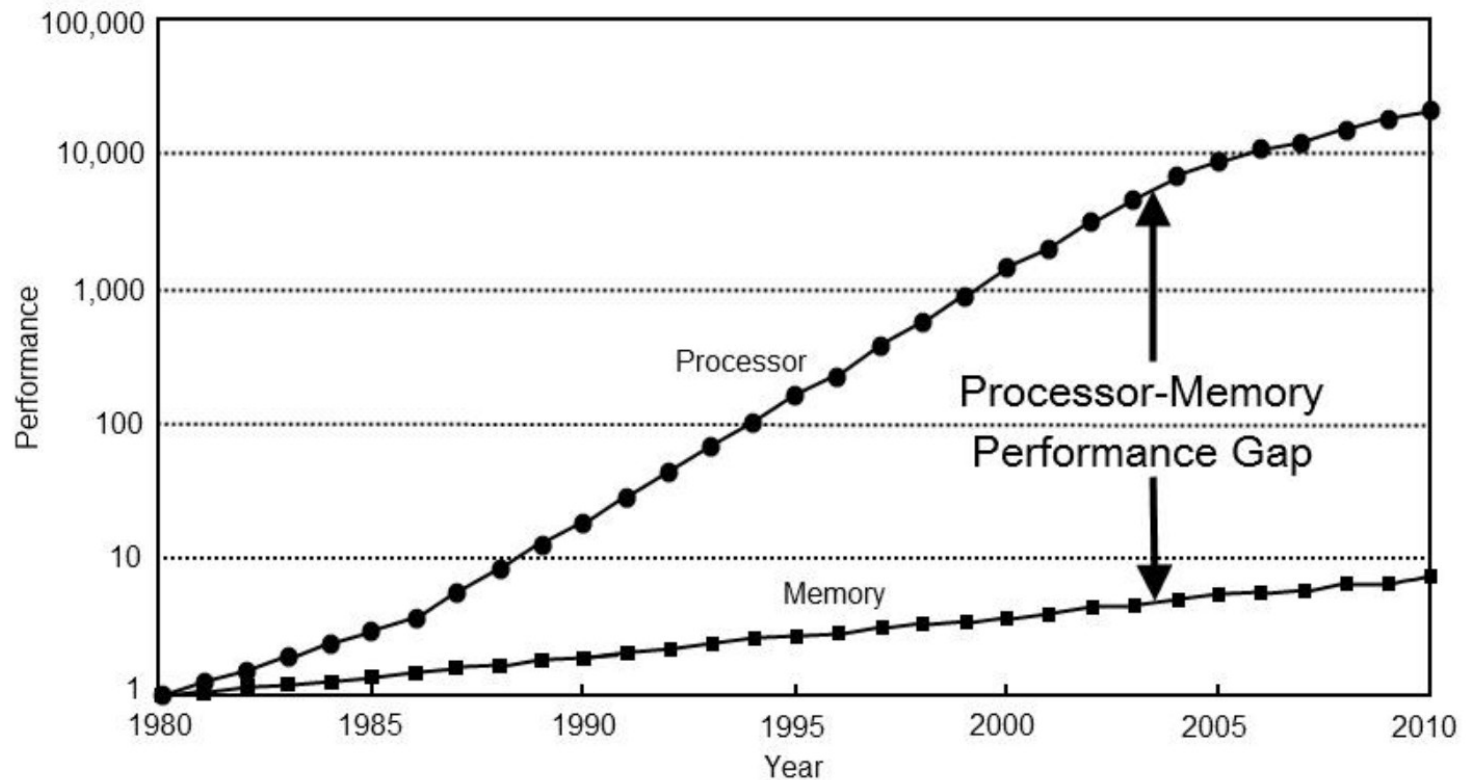


Chapter 5

Large and Fast: Exploiting Memory Hierarchy

Problem - Memory Wall



Computer Architecture: A Quantitative Approach by John L. Hennessy, David A. Patterson, Andrea C. Arpaci-Dusseau

Principle of Locality

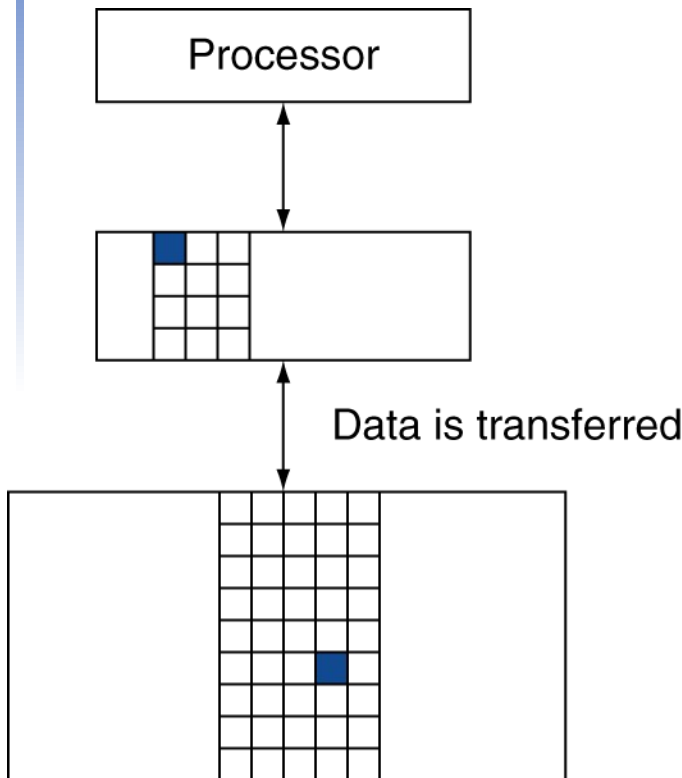
- Programs access a small proportion of their address space at any time
- Temporal locality
 - Items accessed recently are likely to be accessed again soon
 - e.g., instructions in a loop, induction variables
- Spatial locality
 - Items near those accessed recently are likely to be accessed soon
 - E.g., sequential instruction access, array data

Taking Advantage of Locality

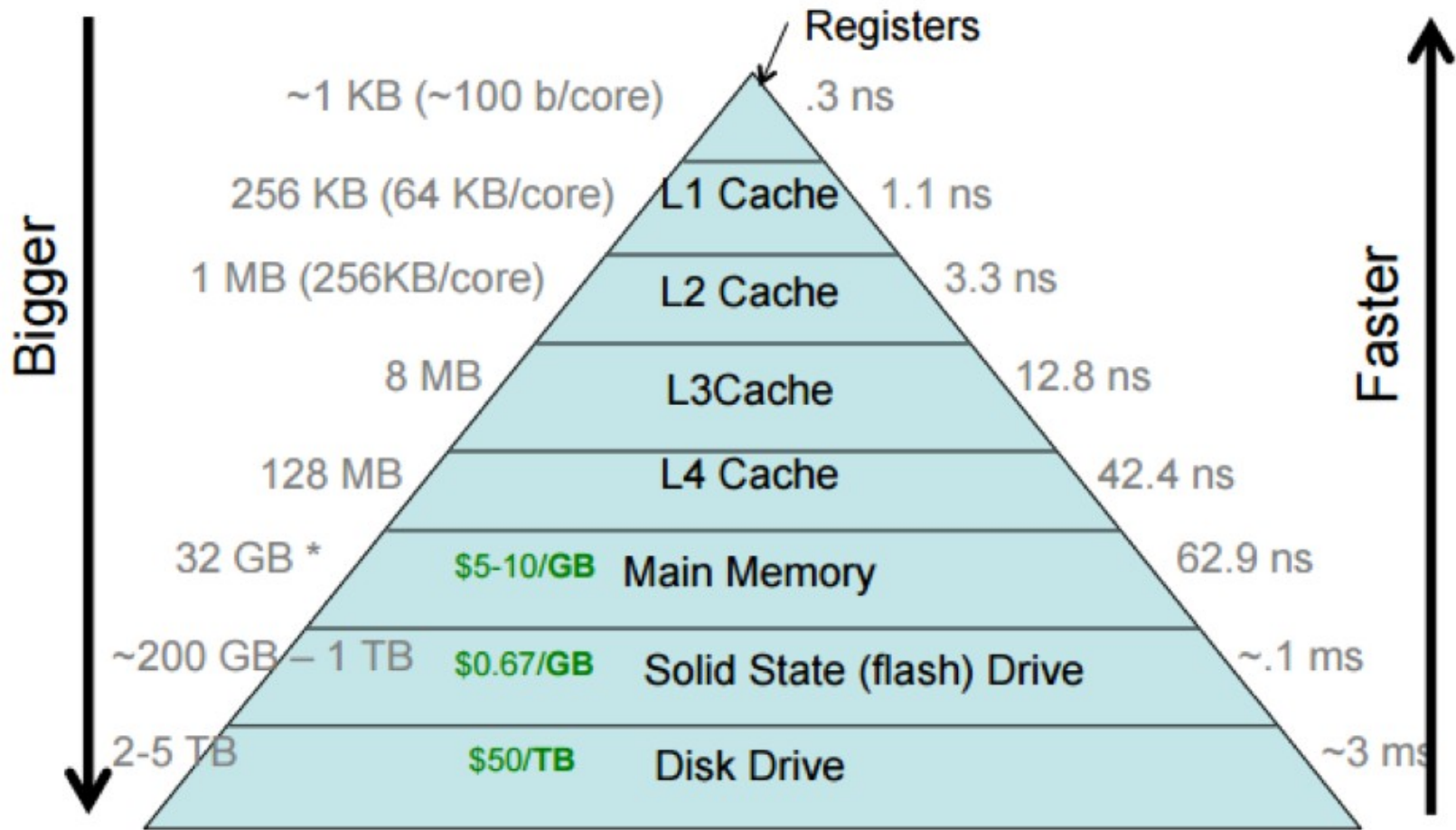
- Memory hierarchy
- Store everything on disk
- Copy recently accessed (and nearby) items from disk to smaller DRAM memory
 - Main memory
- Copy more recently accessed (and nearby) items from DRAM to smaller SRAM memory
 - Cache memory attached to CPU

Memory Hierarchy Levels

- Block (aka line): unit of copying
 - May be multiple words
- If accessed data is present in upper level
 - Hit: access satisfied by upper level
 - Hit ratio: hits/accesses
- If accessed data is absent
 - Miss: block copied from lower level
 - Time taken: miss penalty
 - Miss ratio: misses/accesses
 $= 1 - \text{hit ratio}$
 - Then accessed data supplied from upper level



Memory Hierarchy Levels

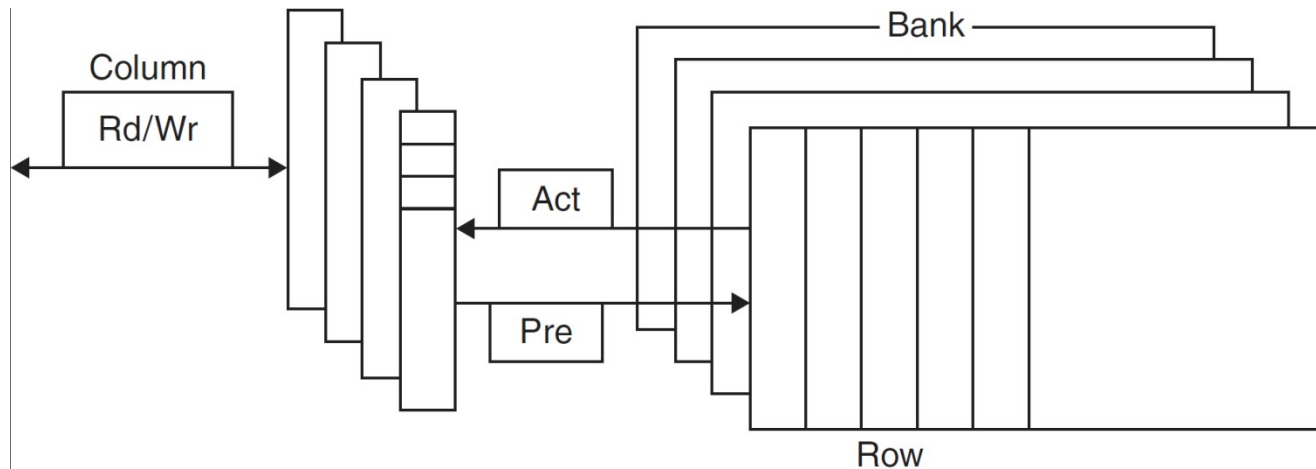


Memory Technology

- Static RAM (SRAM)
 - 0.5ns – 2.5ns, \$2000 – \$5000 per GB
- Dynamic RAM (DRAM)
 - 50ns – 70ns, \$20 – \$75 per GB
- Magnetic disk
 - 5ms – 20ms, \$0.20 – \$2 per GB
- Ideal memory
 - Access time of SRAM
 - Capacity and cost/GB of disk

DRAM Technology

- Data stored as a charge in a capacitor
 - Single transistor used to access the charge
 - Must periodically be refreshed
 - Read contents and write back
 - Performed on a DRAM “row”

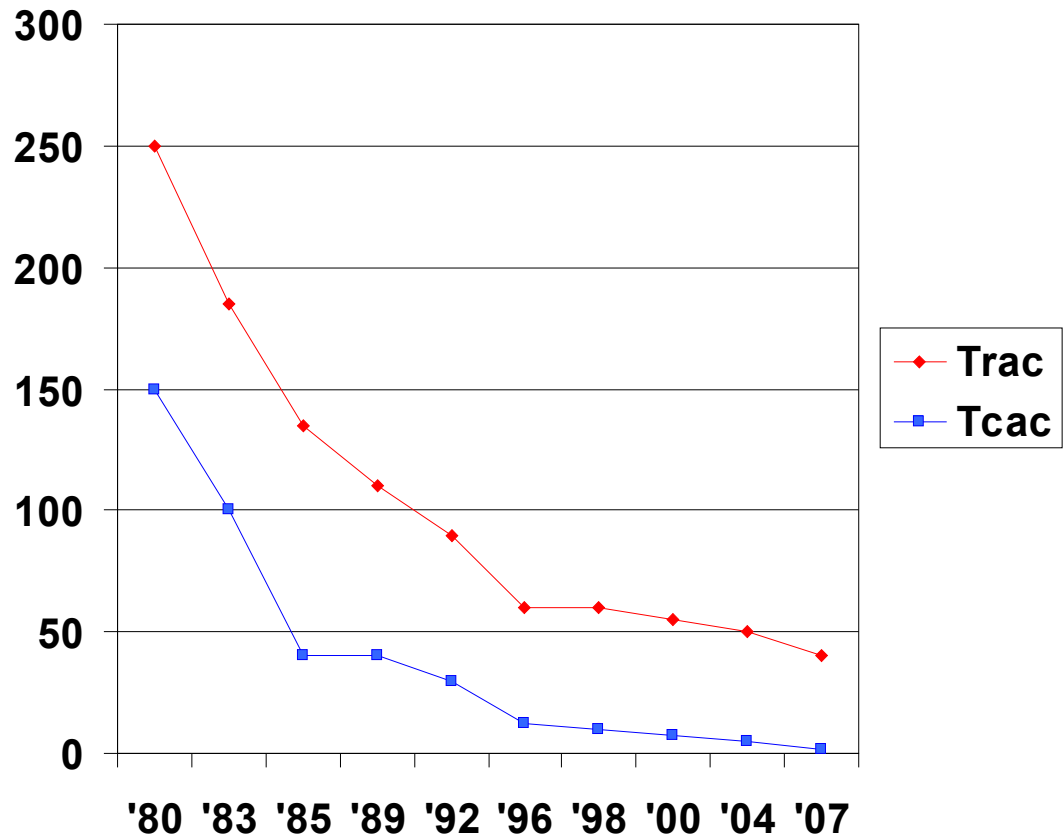


Organization

- Bits in a DRAM are organized as a rectangular array
 - DRAM accesses an entire row
 - Burst mode: supply successive words from a row with reduced latency
- Double data rate (DDR) DRAM
 - Transfer on rising and falling clock edges
- Quad data rate (QDR) DRAM
 - Separate DDR inputs and outputs

DRAM Generations

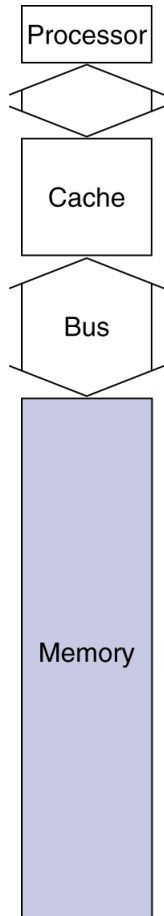
Year	Capacity	\$/GB
1980	64Kbit	\$1500000
1983	256Kbit	\$500000
1985	1Mbit	\$200000
1989	4Mbit	\$50000
1992	16Mbit	\$15000
1996	64Mbit	\$10000
1998	128Mbit	\$4000
2000	256Mbit	\$1000
2004	512Mbit	\$250
2007	1Gbit	\$50



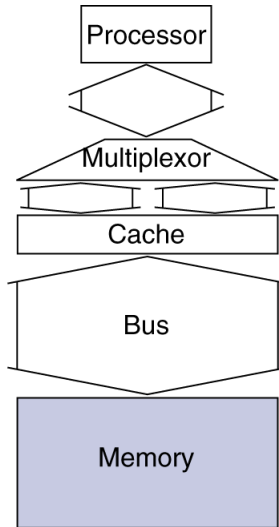
DRAM Performance Factors

- Row buffer
 - Allows several words to be read and refreshed in parallel
- Synchronous DRAM
 - Allows for consecutive accesses in bursts without needing to send each address
 - Improves bandwidth
- DRAM banking
 - Allows simultaneous access to multiple DRAMs
 - Improves bandwidth

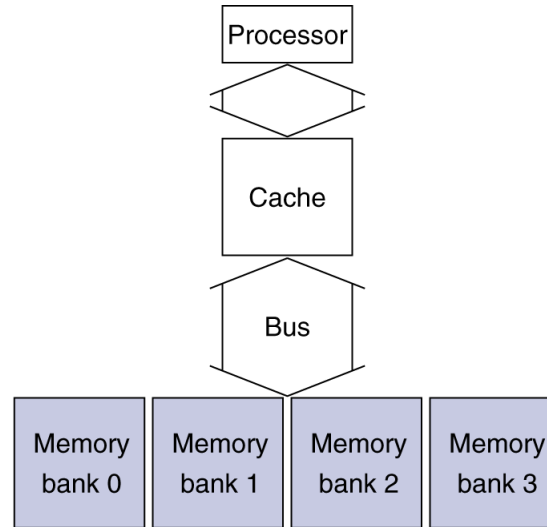
Increasing Memory Bandwidth



a. One-word-wide memory organization



b. Wider memory organization

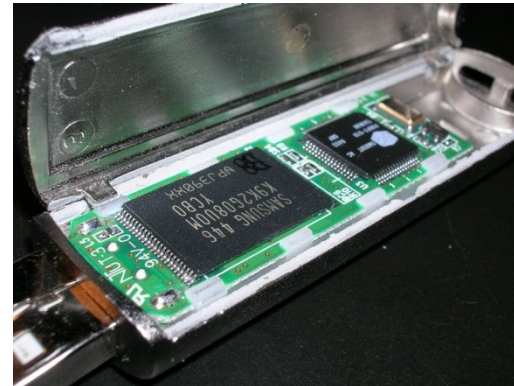


c. Interleaved memory organization

- 4-word wide memory
 - Miss penalty = $1 + 15 + 1 = 17$ bus cycles
 - Bandwidth = $16 \text{ bytes} / 17 \text{ cycles} = 0.94 \text{ B/cycle}$
- 4-bank interleaved memory
 - Miss penalty = $1 + 15 + 4 \times 1 = 20$ bus cycles
 - Bandwidth = $16 \text{ bytes} / 20 \text{ cycles} = 0.8 \text{ B/cycle}$

Flash Storage

- Nonvolatile semiconductor storage
 - 100× – 1000× faster than disk
 - Smaller, lower power, more robust
 - But more \$/GB (between disk and DRAM)

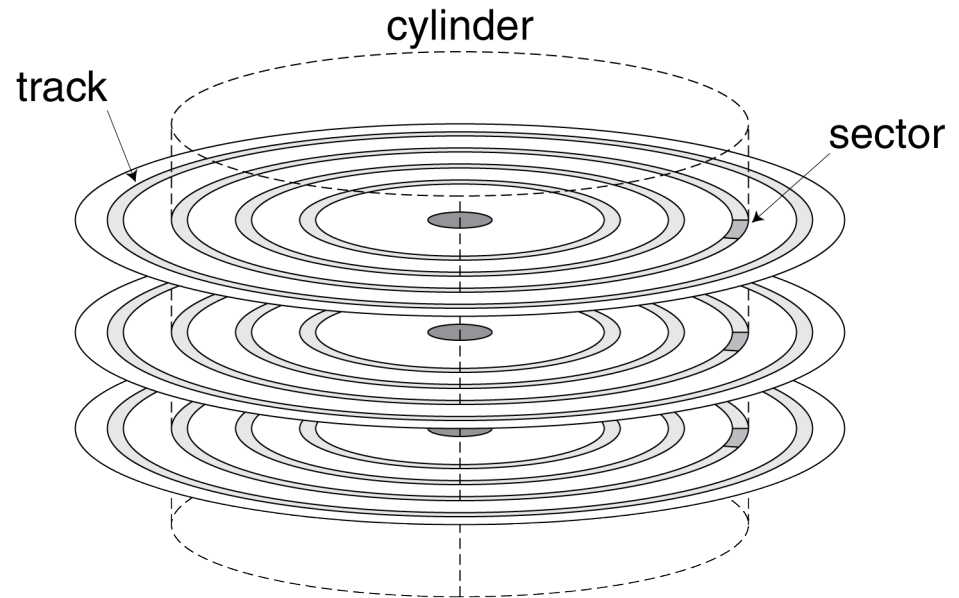


Flash Types

- NOR flash: bit cell like a NOR gate
 - Random read/write access
 - Used for instruction memory in embedded systems
- NAND flash: bit cell like a NAND gate
 - Denser (bits/area), but block-at-a-time access
 - Cheaper per GB
 - Used for USB keys, media storage, ...
- Flash bits wears out after 1000's of accesses
 - Not suitable for direct RAM or disk replacement
 - Wear leveling: remap data to less used blocks

Disk Storage

- Nonvolatile, rotating magnetic storage



Disk Sectors and Access

- Each sector records
 - Sector ID
 - Data (512 bytes, 4096 bytes proposed)
 - Error correcting code (ECC)
 - Used to hide defects and recording errors
 - Synchronization fields and gaps
- Access to a sector involves
 - Queuing delay if other accesses are pending
 - Seek: move the heads
 - Rotational latency
 - Data transfer
 - Controller overhead

Disk Access Example

- Given
 - 512B sector, 15,000rpm, 4ms average seek time, 100MB/s transfer rate, 0.2ms controller overhead, idle disk
- Average read time
 - 4ms seek time
 - + $\frac{1}{2} / (15,000/60) = 2\text{ms}$ rotational latency
 - + $512 / 100\text{MB/s} = 0.005\text{ms}$ transfer time
 - + 0.2ms controller delay
 - = 6.2ms
- If actual average seek time is 1ms
 - Average read time = 3.2ms

Disk Performance Issues

- Manufacturers quote average seek time
 - Based on all possible seeks
 - Locality and OS scheduling lead to smaller actual average seek times
- Smart disk controller allocate physical sectors on disk
 - Present logical sector interface to host
 - SCSI, ATA, SATA
- Disk drives include caches
 - Prefetch sectors in anticipation of access
 - Avoid seek and rotational delay